STAT 236 Course Project Guidelines

Fall 2026

General Information

The final project is a **group assignment**.

- Each group may have **up to 4 students**.
- Possible project formats include:
  - a literature review on a particular research problem;
  - implementation and comparison of existing methods;
  - a new application of existing methods to real data;
  - an extension or improvement of existing methodology or theory; or
  - development of an agent for a statistical task.

  A list of project topics from previous years is provided at the end of this document.
- The project should be related to topics covered in this course. The project should not substantially overlap with the students' existing research projects.

Required Project Components

Each group should submit the following:

(a) **Project report.** The report should describe the research problem, methodology, main results, and relevant technical or empirical details.

(b) **Demonstration webpage.** Each group should create a webpage showcasing the project. All project webpages will be displayed together on the course website.

(c) **Optional presentation.** Students will have the option to present their work in a short lightning talk.

The project grade will be based primarily on the written report. The report accounts for **90%** of the project grade, and the webpage demonstration accounts for **10%**. The optional lightning talk is not graded. More details are given in the grading rubric below.

Student Voting

Each student will rank all project webpages except that of their own group. The m groups receiving the highest overall rankings will receive bonus points, where m will be determined later. Student voting is intended to recognize particularly effective project demonstrations and does not replace the instructor's evaluation of the written report.

Project Timelines

To distribute the project workload, there will be two possible timelines.

Timeline 1

- **Oct. 5 (Monday):** Deadline to form groups and submit a proposed project topic.
- **Nov. 8 (Sunday):** Deadline to submit the report and webpage.
- **Nov. 11:** Optional project presentations.

Timeline 2

- **Nov. 9 (Monday):** Deadline to form groups and submit a proposed project topic.
- **Dec. 6 (Sunday):** Deadline to submit the report and webpage.
- **Reading Period:** Optional project presentations.

Volunteers who would like to follow Timeline 1 are accepted until **Sept. 18 (Friday)**. Those volunteers will automatically be assigned to Timeline 1. The remaining students will be randomly assigned, and the results will be announced on **Sept. 21 (Monday)**.

Instructions for the Report, Webpage, and Presentation

Project Report

There is no fixed template for the report. You are free to choose the report structure that best suits your project. One possible structure is:

- **Introduction:** Describe the research problem and provide relevant background and motivation. If the project involves a data set, briefly introduce the data and explain why it is appropriate for the problem.
- **Methodology:** Describe the statistical model, methodology, or algorithms used in the project. If substantial effort is devoted to data collection, preprocessing, or cleaning, those steps may also be described here.
- **Results:** Present the main empirical or theoretical results. Exploratory data analysis (EDA), including summary statistics or preliminary visualizations, may also be included when relevant.
- **Discussion:** Summarize the main findings, discuss strengths and limitations of the methods, and suggest possible directions for future work.
- **Appendix:** The appendix may include implementation details, supplementary figures and tables, additional simulation results, or other material that is useful but not essential to the main narrative.

Code and implementation details. In general, specific code should **not** be included in the main report. Code should instead be provided as supplementary material when appropriate. However, the report should contain enough methodological and implementation detail for the reader to understand how the analysis was conducted.

For example, it is not sufficient to write only: *"We apply Topic-SCORE (or LDA or NMF) to conduct topic modeling using code downloaded from the following link."* You

should also explain important implementation choices, such as how the number of topics was selected, how tuning parameters were chosen, whether package defaults were used, and whether domain knowledge or other criteria were used to determine parameter values. If default settings from a software package are used, it is sufficient to state this clearly.

Length and formatting. There is no strict page requirement. I recommend using a 12pt font and single line spacing, with the main body of the report, excluding the appendix, kept to approximately **8 pages or fewer**.

Demonstration Webpage

The webpage is intended primarily to **showcase the project to the class**. It does not need to contain the same level of technical detail as the report. There is no required format. Depending on the project, the webpage may include a short description of the problem, the main methodology, key figures or visualizations, major findings, and links to demonstrations or other relevant materials.

You are encouraged to use AI tools to assist with webpage development if useful. Please do **not** spend excessive time polishing the webpage. The written report remains the primary basis for project evaluation.

Optional Lightning Talks

Two class sessions will be reserved for students to present and showcase their projects. Presentations will take the form of **lightning talks**, with each talk limited to **3 minutes**. The goal is to briefly communicate (i) the problem being studied, (ii) the main idea or methodology, and (iii) the most interesting result or takeaway.

Grading Rubric

The rubric below is intended to clarify the main criteria used to evaluate the project. The exact emphasis within each category may vary slightly depending on the nature of the project. For example, a literature-review project and a methodology-development project will naturally demonstrate technical depth in different ways.

Notes on grading.

- Different project types will be evaluated according to the goals of the project. A successful literature review, for example, should demonstrate careful synthesis and critical comparison of the literature, while a methodological project should demonstrate technical understanding and a meaningful analysis of the proposed or modified method.
- Novelty is welcome but is **not required**. A careful, technically sound, and insightful implementation or comparison of existing methods can receive a very strong grade.
- The quality of the statistical reasoning and analysis matters more than the visual polish of the webpage.
- The optional lightning talk is intended for sharing and discussion and does not directly affect the project grade.

Table 1: The grading rubic.

Criterion	Weight	What will be evaluated
Problem formulation and motivation	10%	Clarity and importance of the research question; appropriate background and motivation; clear description of the data or setting when relevant.
Methodology and technical understanding	25%	Correct understanding and use of statistical methods or theory; appropriate methodological choices; explanation of implementation details, tuning parameters, assumptions, and limitations.
Results and analysis	25%	Quality and completeness of the empirical or theoretical analysis; appropriateness of experiments, comparisons, simulations, or proofs; effective use of tables and figures; correctness of conclusions drawn from the results.
Intellectual contribution and insight	15%	Evidence of thoughtful synthesis, interpretation, or extension beyond mechanical implementation; depth of discussion; meaningful comparison of methods or ideas; insight into strengths, limitations, and possible future directions.
Organization and clarity of the report	10%	Logical organization, clear exposition, effective presentation of mathematical and statistical ideas, and overall readability of the report.
Reproducibility and supporting materials	5%	Sufficient information to understand and, when applicable, reproduce the analysis; appropriate documentation of data sources, software, implementation choices, and supplementary code or materials.
Webpage demonstration	10%	Clear and engaging summary of the project; effective communication of the main problem, method, and findings; appropriate use of figures, visualizations, or demonstrations.
Total	**100%**	

Past Project Topics

The following are project topics from recent offerings of STAT 236. They are included as examples to illustrate the range of feasible projects; students are encouraged to develop their own topics.

2024

- Comprehensive Review and Mathematical Analysis of Election Forecasting Models
- Topic Modeling of Song Lyrics
- Information Theoretic Measures for Community Detection
- Using spectral clustering to prune Convolutional Neural Networks
- Cultural Change Among r/lgbt Subreddit Users, 2008-2018
- Community Detection and Network Analysis in Death Penalty Supreme Court Cases
- Network Analysis of 20th Century Newspaper Reprinting

- Inferring advertising provenance from in-app behavior using spectral clustering and SCORE
- Bipartisanship in Congressional Voting Networks
- Persistence in Coauthorship Networks: Evidence from MADStat
- Community Detection in Game of Thrones
- Academic Nationalism
- Product Similarity Network Analysis
- FMRI Spatial Smoothing Using Influential Features Selection
- Political Subreddit Network Analysis
- Text Classification: Identifying Scope Expansion of International Organizations using Job Postings
- Community Detection in Lithuanian Parliamentary Election Data
- Harvard Syllabi: Topic Trends across Harvard STEM curricula
- Gender Discourse on Truth Social: A Computational Analysis of Transgender Narratives
- Dynamic Hierarchical Pitman-Yor Process: A Bayesian Nonparametric Approach to Modeling Evolving Consumer Preferences
- Political Subreddit Network Analysis
- Community Detection on Social Network Datasets
- Mixed Membership in Social Networks: Evaluating Detection Methods with Real-World Data

2023

- Recommender system with link predictions
- Clustering of flower images by network community detection
- Domain Invariant Topic Modeling
- Random graph models and community detection
- Understanding the dynamics of innovation frontier via topic modeling of U.S. patent data
- Kernel-based mixed membership clustering for empirical distributions
- Literature review of algorithms on high-dimensional PCA
- Network analysis using the public comments and ex parte data from the FCC
- Applications and comparisons of community detection methods on covid 19 data
- Network analysis with additional text information: Pubmed as an example
- Literature review of multilayer (multiplex) network analysis
- Community detection to financial market analysis
- Using the Higher Criticism to detect the optimal rank in low-rank models

2022

- Experiments With Competition-Induced Sparsity in Congressional Latent Space Network Models.
- On Generalization and Uncertainty of the Neural Linear Model through the Lens of Effective Dimensionality.
- High-Dimensional Distributed Learning for Causal Inference.
- Siloization in Internet Communities: Lessons from Network Science and Topic Modeling.
- Spectral Clustering for Integrated Data.
- Community Detection And Topic Modeling For Imageclustering In The Leeds Butterfly Dataset.
- Recent Advances in Graph Prediction.
- Belief Propagation and Spectral Redemption in Community Detection.
- BSCORE: An Extension on SCORE.